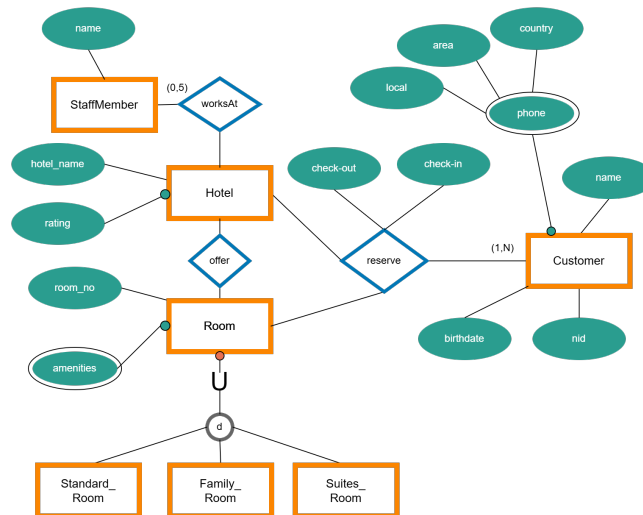


Scientific Data Management & Knowledge Graphs (Winter 2024/25)

Homework 9: Knowledge Graph Creation & Enrichment



Getting started:

- 1) Different from the previous homework, there are 30 points to gain.
- 2) Download the file *data-homework-9.zip* from Stud.IP
- 3) For detailed explanations on how to use the tools in the pipeline, check the slides of Exercise 9.
- 4) Before execution of tools, please check the syntax and mappings files are correctly filled up.
- 5) Complete the blanks in both the mapping and constraints files, then initiate the Docker container.

Task I: RML Mappings

Using the EER diagram above (as a reference for the classes and attributes) and the data sources (see the files in *data/sources/homework*), fill in the blanks in the following RML mapping (and in your file *data/mappings/h9.ttl*).

```
@prefix rr: <http://www.w3.org/ns/r2rml#> .
@prefix ex: <http://example.org/hotel#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix rml: <http://semweb.mmlab.be/ns/rml#> .
@prefix ql: <http://semweb.mmlab.be/ns/ql#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
```

```
<TriplesMapCustomerPhone>
  rml:logicalSource [
    rml:source "/data/sources/homework/CustomerPhone.csv";
    rml:referenceFormulation ql:CSV ];
  rr:subjectMap [
    rr:template "http://example.org/hotel#Customer_{nid}";
    rr:class ex:CustomerPhone ];
  rr:predicateObjectMap [
    rr:predicate ex:country;
    rr:objectMap [
      rml:reference "country"; rr:datatype xsd:int
    ]
  ],
[
```

```

        rr:predicate ex:area;
        rr:objectMap [
            _____; rr:datatype xsd:int
        ]
    ],
    [
        rr:predicate ex:local;
        rr:objectMap [
            rml:reference "local"; rr:datatype xsd:int
        ]
    ] .

<TriplesMapCustomer>
    rml:logicalSource [ _____ ];
    rml:referenceFormulation ql:CSV ];
    rr:subjectMap [
        rr:template "http://example.org/hotel#Customer_{nid}"
        rr:class ex:Customer ];
    rr:predicateObjectMap [
        rr:predicate ex:nid;
        rr:objectMap [
            rml:reference "nid";
            rr:datatype xsd:int
        ]
    ],
    [
        rr:predicate ex:name;
        rr:objectMap [
            rml:reference "name";
            rr:datatype xsd:string
        ]
    ],
    [
        rr:predicate ex:birthdate;
        rr:objectMap [
            rml:reference "birthdate";
            rr:datatype xsd:date
        ]
    ],
    [
        rr:predicate ex:hasPhoneNumber;
        rr:objectMap [
            _____ <TriplesMapCustomerPhone>;
            _____ [rr:child _____; rr:parent "nid"]
        ]
    ] .

<TriplesMapHotel>
    rml:logicalSource [
        rml:source "/data/sources/homework/Hotel.csv";
        rml:referenceFormulation ql:CSV ];

    rr:subjectMap [
        rr:template "http://example.org/hotel#Hotel_{ID}";
        rr:class ex:Hotel ];
    rr:predicateObjectMap [
        rr:predicate ex:name;
        rr:objectMap [
            rml:reference "name";
            rr:datatype xsd:string
        ]
    ],
    [
        rr:predicate ex:rating;
        rr:objectMap [
            rml:reference "rating";
            rr:datatype xsd:int
        ]
    ] .

<TriplesMapReservation>
    rml:logicalSource [
        rml:source "/data/sources/homework/Reservation.csv";

```

```

rml:referenceFormulation ql:CSV ];

rr:subjectMap [
  rr:template "http://example.org/hotel#Reservation_{ID}";
  rr:class ex:Reservation ];
rr:predicateObjectMap [
  rr:predicate ex:reservedHotel;
  rr:objectMap [
    rr:template "http://example.org/hotel#Hotel_{Hotel}"
  ]
],
[
  rr:predicate ex:reservedRoom;
  rr:objectMap [
    rr:template "http://example.org/hotel#Room_{Room}"
  ]
],
[
  rr:predicate ex:reservedCustomer;
  rr:template "http://example.org/hotel#Customer_{Customer}"
],
[
  rr:predicate ex:checkIn;
  rr:objectMap [
    rml:reference "check-in";
    rr:datatype xsd:date
  ]
], [
  rr:predicate ex:checkOut;
  rr:objectMap [
    rml:reference "check-out";
    rr:datatype xsd:date
  ]
] .

```

```

<TriplesMapRoom>
  rml:logicalSource [
    rml:source "/data/sources/homework/Room.csv";
    rml:referenceFormulation ql:CSV ];

  rr:subjectMap [ rr:template "http://example.org/hotel#Room_{no}";
    rr:class ex:Room ];
  rr:predicate ex:roomNo;
  rr:objectMap [
    rml:reference "no"; rr:datatype xsd:int
  ]
], [
  rr:predicate rdf:type;
  rr:objectMap [
    rr:template "http://example.org/hotel#Room_{type}"
  ]
] .

```

```

<TriplesMapAmenity>
  rml:logicalSource [
    rml:source "/data/sources/homework/Amenities.csv";
    rml:referenceFormulation ql:CSV ];
  rr:subjectMap [
    rr:template "http://example.org/hotel#Room_{room}";
    rr:class ex:Room ];
  rr:predicateObjectMap [
    rr:predicate ex:hasAmenity;
    rr:objectMap [
      rml:reference "amenity"; rr:datatype xsd:string
    ]
  ]
] .

```

```

<TriplesMapStaffMember>
  rml:logicalSource [
    rml:source "/data/sources/homework/StaffMember.csv";
    rml:referenceFormulation ql:CSV ];
  rr:subjectMap [

```

```

rr:template "http://example.org/hotel#Staff_{ID}";
rr:class ex:StaffMember ];
rr:predicateObjectMap [
  rr:predicate ex:staffName;
  rr:objectMap [
    rml:reference "name";
    r:datatype xsd:string
  ]
] .

<TriplesMapWorkContracts>
  rml:logicalSource [
    rml:source "/data/sources/homework/WorkContracts.csv";
    ] ;
  rr:subjectMap [
    rr:template "http://example.org/hotel#Staff_{Staff}";
    ex:StaffMember ];
  rr:predicateObjectMap [
    rr:predicate ex:worksAt;
    rr:objectMap [
      rr:template "http://example.org/hotel#Hotel_{Hotel}"
    ]
  ]
] .

```

Task II: SPARQL Queries over the Data

Write the following SPARQL queries (repository *homework9_provided_data* in GraphDB [see Exercise 9]) and **provide the query as well as the answer**:

- List the names of all staff members.
- Get the name of the hotel, the customer name and NID, and the room assigned to the customer respectively.
- For each person with a NID less than 20,000,000 return its URI, NID, and birthdate.

Task III: SPARQL Queries over the Ontology

Write the following SPARQL queries (repository *homework9_ontology* in GraphDB [see Exercise 9]) and provide the query as well as the answer:

- How many classes (RDFS) are in the ontology?
- For the properties with the range *Hotel*, return the properties and their domain.
- For each class in the domain of a property, return the class, the number of datatype properties, and the number of object properties. Order the answer like the following: Lower number of datatype properties first, if they are the same, lower number of object properties first.

Task IV: Integrity Constraints with SHACL

- Fill in the blanks in the SHACL constraints (and in your file `data/constraints/homework/homework9.ttl`). For your convenience, we also provide a natural language representation of the integrity constraints.
 - Customers have exactly one name, birth date, and NID.
 - Customers have one or two phone numbers.
 - Staff members have exactly one name.
 - Staff members work at up to five hotels that satisfy the constraints for hotels.
 - Hotels have exactly one name and rating.
 - Rooms have exactly one room number and amenity.
 - Reservations are made by at least one valid customer.
 - Reservations are made for one to two valid rooms.
 - Reservations are made in at least one valid hotel.

10. Reservations have exactly one check-in and check-out date.

```
@prefix ex: <http://example.org/hotel#> .
@prefix exS: <http://example.org/shapes/> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix sh: <http://www.w3.org/ns/shacl#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

exS:CustomerShape a sh:NodeShape ;
  sh:targetClass ex:Customer ;
  sh:property [
    sh:path ex:name ;
    sh:minCount 1 ;
    sh:maxCount 1
  ] ;
  sh:property [
    sh:path ex:birthdate ;
    sh:minCount 1 ;
    sh:maxCount 1
  ] ;
  sh:property [
    sh:path ex:nid ;
    sh:minCount 1 ;
    sh:maxCount 1
  ] ;
  sh:property [
    sh:path ex:hasPhoneNumber ;
    sh:minCount 1 ;
    sh:maxCount 1
  ] .

exS:StaffMemberShape a sh:NodeShape ;
  sh:targetClass ex:StaffMember ;
  sh:property [sh:path ex:staffName; sh:minCount 1; sh:maxCount 1];
  sh:property [
    sh:path ex:worksFor ;
    sh:qualifiedMaxCount 5
  ] .

exS:HotelShape a sh:NodeShape ;
  sh:targetClass ex:Hotel ;
  sh:property [sh:path ex:name; sh:minCount 1; sh:maxCount 1];
  sh:property [sh:path ex:rating; sh:minCount 1; sh:maxCount 1] .

exS:RoomShape a sh:NodeShape ;
  sh:targetClass ex:Room ;
  sh:property [sh:path ex:hasAmenity; sh:minCount 1; sh:maxCount 1];
  sh:property [sh:path ex:hasAmenity; sh:minCount 1; sh:maxCount 1];

exS:ReservationShape a sh:NodeShape ;
  sh:targetClass ex:Reservation;
  sh:property [ sh:path ex:reservedCustomer ; sh:qualifiedValueShape exS:CustomerShape ;
    sh:qualifiedMinCount 1 ];
  sh:property [ sh:path ex:reservedRoom ; sh:qualifiedValueShape exS:RoomShape;
    sh:qualifiedMinCount 1 ; sh:qualifiedMaxCount 2 ];
  sh:property [ sh:path ex:reservedHotel ; sh:qualifiedValueShape exS:HotelShape ;
    sh:qualifiedMinCount 1 ];
  sh:property [sh:path ex:checkIn ; sh:minCount 1; sh:maxCount 1];
  sh:property [sh:path ex:checkOut ; sh:minCount 1; sh:maxCount 1].
```

b) Which natural language statements (one or more) are correct for the given SHACL shape?

- i) exS:StudentShape a sh:NodeShape ;
sh:targetClass ex:Student ;
sh:property [sh:path ex:hasAdvisor ; sh:minCount 1 ; sh:maxCount 2] .
- ☐ The shape defines constraints for entities of class ex:Student.
 - ☐ Valid entities must have at least one advisor.
 - ☐ Valid entities cannot have more than two advisors.
 - ☐ Valid entities must have exactly two advisors.
- ii) exS:DoctorShape a sh:NodeShape ;
sh:targetClass ex:Doctor ;
sh:property [sh:path ex:practicesAt ; sh:node exS:HospitalShape;
sh:minCount 1 ; sh:maxCount 2] .

- ☐ Valid entities cannot occur in the subject of triples with the predicate `ex:practicesAt`.
- ☐ Valid entities must be linked to at least one hospital without validation..
- ☐ Valid entities must be linked to 1-2 hospitals that validate against `exS:HospitalShape`.
- ☐ If a connected hospital fails to validate against `exS:HospitalShape`, the doctor entity becomes invalid.

Task V: Validation

Provide the validation result, i.e., valid or invalid, for the following instances.
You can use *Trav-SHACL* for the validation.

- a) `http://example.org/hotel#Customer_12378457`
- b) `http://example.org/hotel#Staff_5`
- c) `http://example.org/hotel#Hotel_5`
- d) `http://example.org/hotel#Reservation_2`
- e) `http://example.org/hotel#Room_55`

Submission

Please, assign yourself to one of the mentors in Stud.IP (see groups in the exercise). We kindly ask you to distribute yourselves evenly. Submit your solution as a single PDF via e-mail to your mentor until Friday, **January 10, 2025** at **11:00** sharp. Use the subject *[SDMKG Homework] Submission 9*. Please, include your name and matriculation number in the name of the PDF file and on at least the first page of your submission.

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